Chan 2021 SCLC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

24 of 34 malignant-bearing samples (≥37 cells each) intermix on one continuous manifold with no per-patient archipelago (radius-graph: 118/50,999 island cells removed; iLISI median 6.17 of 24); proliferation lobe kept — modules are cell-cycle-free.

Gate 2 — module geography → GO (non-branching)

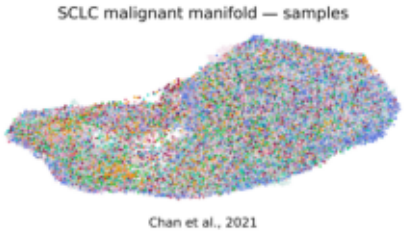
Single embryonic axis: Gast and pSom co-vary (smoothed r=+0.48 — overlap, not competitive cross-repression); Crest is an inverse terminal gradient rising where pSom falls (pSom↔Crest r=−0.80). No hub diverging into two arms; the Gast↔pSom co-variation strengthens in non-cycling cells (+0.48→+0.59).

Gate 3 — cross-repression → NA

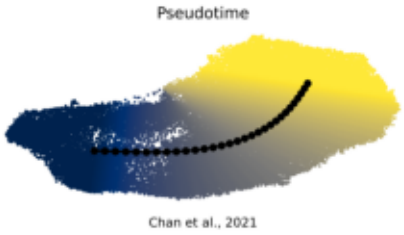
Non-branching manifold — a curve with 2 tips / 0 forks (PPT tree also 0 forks), so there is no branch pair and no convergence hub to test for cross-repression.

STEP FIGURES

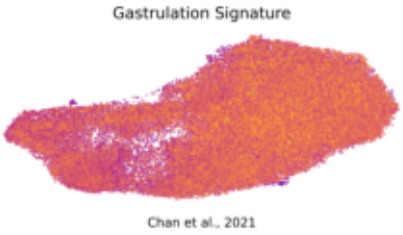
Manifold — samples



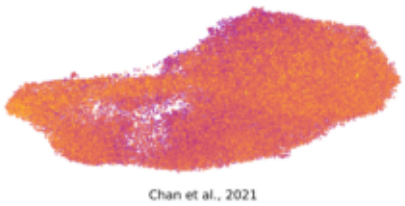
Pseudotime



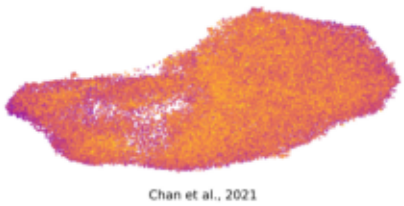
Gastrulation



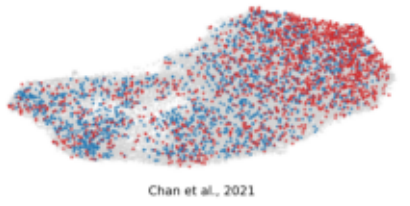
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

SCLC (Chan et al., 2021), 10x — 37 samples / 86,662 cells (CCCA master scope). Malignant only: 51,052 cells across 34 samples → min-cell threshold (≥37 cells/sample) keeps 50,999 cells across 24 samples → island removal → 50,881 cells enter the trajectory; whole manifold (no lobe scoping).

Verdict.

Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Chan's SCLC malignant manifold is a single continuous gastrulation/somitogenesis → terminal-neural-crest gradient rooted at the Gast/pSom-high pole, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. SCLC-A/SCLC-N subtypes do not form two module-opposed arms, and the embedding carries a proliferation axis, but the embryonic modules are cell-cycle-free (|r| with cycling ≤0.16) and the Gast↔pSom co-variation strengthens in non-cycling cells (+0.48→+0.59; the pSom↔Crest anti-correlation persists, −0.80→−0.77), so the whole manifold is kept. The curve is fold-free (far-node approach / edge = 24.7). EMT-TFs (widened panel incl. FOXC/PRRX): TWIST1 (root-leaning) → SNAI1 (terminus-leaning), all weakly pseudotime-associated (|ρ|≤0.11) — consistent with a diffuse linear axis.

Parameter choices

Min cells/sample	≥37 (dominant 92→15 cliff)	**Integration**	Harmony θ=5; n_pcs=20
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=30, λ=0.3, μ=100
Root node	tip 2 = Gast/pSom-high	**Overdispersed HVG**	3,228
Geometry	Linear (non-branching); 2 tips / 0 forks	**EMT-TFs**	TWIST1 (root) · SNAI1 (terminus); |ρ|≤0.11